

Testing the Transferability of Critical-Satellite Rankings in LEO Networks

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Abstract—A list of critical satellites is usually produced under a particular routing policy. Can that list still identify damaging failures after the policy changes? Different routes naturally expose different satellites, but exposure alone does not answer this question. We evaluate ranking transfer through single-satellite deletion, using a common evaluation universe and comparing each flow with its own no-failure baseline. Damage Capture measures how much target-policy deletion damage a reused ranking captures relative to a target-policy oracle; Transfer Regret measures the shortfall. In a controlled 1,156-satellite Kuiper-style shell, shortest-distance and hop-count damage rankings transfer well, despite differences in exposure. A utilization-aware link-state policy yields a narrower exception: its peak-link-stress ranking transfers poorly to IGP-ECMP. Across 24 sampled orbital phases, median top-20 Transfer Regret is 0.248, with a traffic-cluster bootstrap interval of [0.150, 0.333], and the direction passes a six-phase, three-hash confirmation criterion. The corresponding shortest-distance direction fails that confirmation. Expanding ISL density and GSL attachment multiplicity does not resolve the transfer loss in the tested settings. Pre-deletion features also fail our held-out prediction criterion. These findings support testing the reuse of failure rankings by damage dimension, routing state, and orbital phase, rather than inferring general criticality changes from exposure alone. They concern graph-level failure effects, not measured service latency or an operational routing recommendation.

Index Terms—LEO satellite networks, resilience, routing, failure impact, ranking transfer

I. INTRODUCTION

A satellite failure matters because of the service that depends on it. In a low Earth orbit (LEO) network, that dependence changes as satellites move, ground access changes, and routing distributes traffic across available paths. Topology design and route variability have therefore received considerable attention [1], [2]. Resilience studies also identify heavily used satellites and test the consequences of removing them [3]. Such analyses produce useful rankings, but every ranking is conditional on a routing configuration.

Suppose an analyst identifies the twenty most damaging single-satellite failures under shortest-distance routing. A different policy is then adopted without changing the topology or traffic demands. Should the same twenty satellites remain the priority for failure testing? It is easy to show that the policies use different paths. The more useful question is whether reusing the original list misses a substantial fraction of the failures that matter under the new policy. Two lists may have little overlap yet capture nearly the same damage, particularly when many satellites have similar scores. Conversely, modest path changes may alter which removals create a bottleneck.

We study this reuse problem directly. For a fixed graph, traffic instance, and policy state, we remove each satellite in a common policy-dependency universe, reroute the same flows, and record the resulting damage. The source ranking and the evaluation policy are then varied separately. The comparison is against the target policy’s own deletion-defined ranking, not against its exposure ranking. This distinction separates the quality of a predictor from the transferability of failure importance itself.

The dependence of network risk on traffic and routing is not, by itself, a new observation. Load-dependent analyses such as HYDRA already challenge purely topological notions of importance [4]. Our contribution is a way to quantify the cost of reusing a ranking across routing policies and to test where that cost persists. We call the fraction of target-oracle damage recovered by a reused top- K list *Damage Capture*; its complement is *Transfer Regret*. Both are specific to a damage dimension. Losing reachable demand, increasing propagation distance, and concentrating load need not identify the same satellites.

We ask three questions. First, does deletion-defined criticality transfer between routing policies, rather than merely changing with exposure? Second, does a transfer loss persist across orbital phases and ECMP hash choices? Third, do controlled changes in traffic or available links explain or reduce that loss? We use a Hypatia-derived graph pipeline [5], shortest-distance routing, integer-metric IGP-ECMP, and a utilization-aware link-state model. Hop-count routing and a two-pass stress-aware probe provide diagnostic comparisons. These are controlled routing models, not implementations of an operator’s control plane.

The results are more selective than an exposure-only analysis suggests. Shortest-distance and hop-count deletion rankings transfer well. Utilization-aware rankings have a material transfer loss to shortest-distance and IGP-ECMP in the short-window experiment, but only for peak-link-stress damage. Over an orbit, only the IGP-ECMP evaluation direction meets the complete confirmation criterion. A denser ISL/GSL construction does not restore transfer in the tested comparison, and our pre-deletion features do not predict regret reliably enough under held-out traffic seeds. These negative results delimit the claim: routing can change which satellites are important for a particular failure effect, but our evidence does not establish universal ranking instability or a general explanation for it.

The paper contributes a deletion-based transfer protocol with explicit treatment of policy dependencies, flow pairing,

ties, and zero-damage cases; a controlled evaluation across routing policies and sampled orbital phases; and traffic and topology experiments that test, rather than assume, explanations for the observed transfer loss.

II. NETWORK AND ROUTING MODEL

A. Graphs, Traffic, and Exposure

At time t , the graph $G_t = (V_t, E_t)$ contains satellites and ground stations, with inter-satellite links (ISLs) and ground-to-satellite links (GSLs). Edge weight $w(e)$ is geometric distance in meters. A route joins two ground stations through satellites. Other ground stations are excluded from that flow's routing graph, so they cannot act as terrestrial relays. The current graph is undirected: traffic in either direction contributes to the same edge-load total. It does not model separate transmit and receive capacities.

A scenario fixes the graph, directed source–destination flows, their demand weights, and routing parameters. Every policy sees the same flow instances. Demand d_f is a synthetic offered-load weight, not Mbps. For policy p , its final paths $P_p(f)$ induce link load and stress:

$$\ell_p(e) = \sum_{f: e \in P_p(f)} d_f, \quad \sigma_p(e) = \frac{\ell_p(e)}{C}. \quad (1)$$

All links have capacity proxy $C = 10$ in the same units as demand. Stress is dimensionless; values above one indicate offered load above that proxy, not observed packet loss.

Demand-weighted satellite exposure is

$$E_p(s) = \sum_f d_f \mathbf{1}\{s \in P_p(f)\}. \quad (2)$$

Only reachable flows contribute, each at most once per satellite. Exposure is a pre-deletion predictor. We do not define a satellite as critical merely because it has high exposure, nor interpret low overlap between exposure lists as evidence of high failure damage.

B. Routing Policies and Failure Response

Shortest-distance minimizes total edge distance. The diagnostic **hop-count** policy minimizes edge count, with distance as its tie-breaker.

IGP-ECMP quantizes distance to positive integer metrics at 10-km resolution. A stable per-flow hash chooses among equal-cost next hops. The experiment uses hash seeds 0, 1, and 2 where specified. This models unsplittable flow-based ECMP, not fluid splitting or per-packet balancing. Hash-based next-hop selection is a standard ECMP mechanism [6]; our implementation is a graph-level instance of that mechanism, not a reproduction of a particular router.

Utilization-aware link state (Util-LS) retains IGP-ECMP forwarding but changes advertised metrics using delayed load observations. Each fixed-topology scenario starts with the base IGP metrics and runs 20 state-update intervals. At each interval, the observed link stress updates an exponentially weighted moving average with gain 0.1. The advertised utilization is capped at 2.0 and changes only when its difference from the

previous advertisement reaches 0.1. For base integer metric $m_0(e)$ and advertised utilization $\hat{u}(e)$, the next metric is

$$m(e) = m_0(e) + \lfloor 0.05m_0(e)\hat{u}(e) + 0.5 \rfloor. \quad (3)$$

Updates represent five-second intervals with a one-second observation delay. Since the delay is shorter than the interval, the previous interval's sample is available at the next update. This is a discrete state model, not a simulation of link-state message delivery.

For deletion, Util-LS freezes the final pre-failure advertisements, removes the failed node, and recomputes SPF/ECMP on the surviving graph. It does not obtain post-failure global load to choose new weights. The experiment measures the routing response under those advertisements, not eventual post-failure load-adaptation convergence. Warm-up is repeated independently at each snapshot; state is not carried between orbital phases.

The **stress-aware two-pass probe** first routes by shortest distance, computes first-pass stress, and reroutes using $w'(e) = w(e)(1 + 2\sigma_1(e))$. After a failure it recomputes both passes. It deliberately provides a stronger perturbation and is not presented as an operational traffic-engineering algorithm. We keep its results distinct from those of IGP-ECMP and Util-LS.

III. MEASURING RANKING TRANSFER

A. A Common Deletion Universe

The satellite set used for evaluation must not favor either policy. For scenario x , let $B_p(x)$ contain satellites on which policy p 's baseline decisions depend, and define

$$U_x = \bigcup_{p \in \mathcal{P}} B_p(x), \quad (4)$$

where the policy family $\mathcal{P}$ is fixed for each experiment before its deletion results are examined. Every policy is evaluated on every satellite in U_x .

Final paths alone are insufficient to construct this universe. ECMP dependencies include unselected alternatives in each flow's equal-cost shortest-path DAG. The two-pass probe depends on first-pass paths as well as final paths. Util-LS exposes the equal-cost dependencies under its frozen advertisements. A node absent from final traffic paths may therefore still affect a routing decision. In contrast, a node outside the evaluation policy's complete dependency set is a strict inactive control and should produce zero damage under deterministic replay. This screening makes graph-level evaluation tractable without treating the entire unused constellation as the main random baseline.

For each $s \in U_x$, we delete s and its incident edges, reroute the unchanged flows, and compare against that evaluation policy's own no-deletion baseline. Each intervention removes *one* satellite. Top- K is the budget for selecting separate failure tests, not a simultaneous K -satellite attack. We do not infer multi-failure damage by adding these single-failure results.

B. Damage Dimensions

Let $D_p^m(s)$ be the computed deletion damage under policy p for dimension m . “Deletion-defined” refers to an intervention in the graph model, not a measured real-world failure. We retain separate outcomes rather than combine them into an arbitrary risk score.

For peak link stress, the outcome is

$$D_p^{\text{stress}}(s) = \max_{e \in E_t \setminus \delta(s)} \sigma_{p,-s}(e) - \max_{e \in E_t} \sigma_p(e), \quad (5)$$

where $\delta(s)$ contains the deleted node’s incident edges. Signed values are retained. A negative value means the largest modeled stress decreased, which can arise from rerouting or from lost traffic; it is not automatically a resilience improvement. Lost demand is therefore reported separately. Under uniform capacity, peak stress and peak load are proportional and are not independent confirmations.

For delay-related outcomes, we first pair flows by identity. The propagation RTT estimate of path P is

$$R(P) = \frac{2 \sum_{e \in P} w(e)}{c}, \quad (6)$$

where c is the vacuum speed of light. The factor two assumes the same path length in both directions. Processing, transmission, and queueing delays are absent. Let F_p^0 be baseline-reachable flows and F_p^s those reachable both before and after deleting s . Positive propagation burden is

$$D_p^{\text{prop}}(s) = \frac{\sum_{f \in F_p^s} d_f \max(R_{p,-s}(f) - R_p(f), 0)}{\sum_{f \in F_p^0} d_f}. \quad (7)$$

The lost-demand fraction uses the demand of $F_p^0 \setminus F_p^s$ with the same denominator. Rerouted-demand fraction counts path changes among paired reachable flows, also normalized by baseline-reachable demand. Neither metric silently treats a disconnected flow as a short route.

We also calculate the demand-weighted p95 of per-flow propagation stretch, $R_{p,-s}(f)/R_p(f) - 1$, over F_p^s . It is reported only when effective support $(\sum d_f)^2 / \sum d_f^2$ is at least 20. Unsupported tails are missing, not zero. This differs from subtracting the p95 of two flow populations, which need not measure the change experienced by the same flows.

The implementation retains modeled congestion-adjusted RTT, $R(P)(1 + \beta \max_{e \in P} \sigma_p(e))$, as a secondary diagnostic. It is not packet-level RTT, and its p95 impact is not the principal evidence used here. The main transfer results do not depend on β or on that load-to-delay formula.

C. Damage Capture and Transfer Regret

For a chosen dimension m , let $T_p^m(K)$ be the K satellites with largest deletion damage under source policy p . Write $D_q^{m,+}(s) = \max(D_q^m(s), 0)$. Reusing the source list under target policy q gives

$$\begin{aligned} \text{DC}_{p \rightarrow q}^m(K) &= \frac{\sum_{s \in T_p^m(K)} D_q^{m,+}(s)}{\sum_{s \in T_q^m(K)} D_q^{m,+}(s)}, \\ \text{TR}_{p \rightarrow q}^m(K) &= 1 - \text{DC}_{p \rightarrow q}^m(K). \end{aligned} \quad (8)$$

The denominator is a target-policy oracle within the same evaluation universe. Thus, $\text{TR} = 0.25$ means that the reused list misses one quarter of the summed positive single-failure damage captured by the target’s own top- K list. It does not mean that one quarter of all network damage, traffic, or satellites is lost. The oracle is an offline evaluation bound, not a cheap deployable predictor.

The two lists have the same size and universe, so $0 \leq \text{DC}, \text{TR} \leq 1$. If the target oracle has no positive damage, both are undefined. We report that support explicitly. Diagonal capture equals one by construction and is a normalization check, not a discovery. We use deterministic satellite-ID tie-breaking and repeat ambiguous cutoffs with 1,000 random tie resolutions. The primary comparison uses $K = 20$; K always denotes separate tests.

Exposure selectors are evaluated against the same oracle. Uniform random selectors draw from U_x , rather than from all constellation nodes, using 10,000 sets per scenario. Once the deletion table exists, these are resampling operations and require no additional routing runs. Random sets may overlap the exposure list. This baseline tests selection quality within the policy-dependency universe; it is not matched on structural centrality.

IV. EXPERIMENTAL DESIGN

A. Topology and Evaluation Stages

The controlled shell uses 34 orbital planes with 34 satellites each, nominal altitude 630 km, and inclination 51.9° . These parameters follow the public Kuiper planning appendix [7]. The resulting 1,156-satellite graph is a *Kuiper-style synthetic shell*, not the current deployed network. The fixed synthetic orbital input has mean motion 14.8 revolutions/day, yielding a modeled period of 5,837.84 s. Ground locations are the top 100 populated places in our fixed Natural Earth input [8], not an observed operator gateway deployment.

The base ISLs form a plus-grid with in-plane and adjacent-plane neighbors. Stored geometric cutoffs are 5,442.958 km for ISLs and 1,260 km for GSLs. The base graph includes visible GSLs under that distance rule; the access-density experiment imposes additional nearest-visible attachment limits. Geometry and source files remain fixed within each comparison.

Table I separates the evaluation stages. The short-window and orbit experiments use four synthetic traffic models: equal-weight pairings, population-weighted demand, gravity-style demand, and regional hotspots. Each traffic-model/seed instance has 50 directed flows and is reused across its snapshots. These traffic models are not fitted to measured LEO traffic. The later factorial experiment controls total demand explicitly instead of interpreting the four models as independent mechanism interventions.

The short window samples 0, 50, 100, 150, and 199 s. Orbital coverage uses 24 equally spaced phase bins, rounded to seconds, from 0 to 5,595 s, excluding the duplicate orbit endpoint. Six phases (0, 973, 1,946, 2,919, 3,892, and 4,865 s) also use hashes 1 and 2. These are independently warmed

TABLE I

COMPLETED EVALUATION STAGES. A SCENARIO-STATE FIXES TOPOLOGY, TRAFFIC, SNAPSHOT, AND HASH SEED. A DELETION-POLICY ROW IS ONE SATELLITE REMOVAL UNDER ONE EVALUATION POLICY; ROWS ARE NOT INDEPENDENT STATISTICAL REPLICATES. SEPARATE STAGES REUSE SOME SCENARIOS AND ARE NOT POOLED INTO ONE SAMPLE.

Stage	Controlled scope	States	Deletion rows
Predictors (009)	5 snapshots, 4 traffic models, 3 seeds; shortest, hop-count, stress probe	60	85,338
Policy transfer (010)	Same traffic/snapshot scope, 3 ECMP hashes; shortest, IGP-ECMP, Util-LS, stress probe	180	339,688
Traffic factors (011)	2 ⁴ regimes, 10 seeds, 3 phases; three non-probe policies, hash 0	480	276,495
Orbital phases (012)	24 phases at hash 0; 6 of those phases also at hashes 1/2; four policies	432	822,996
Topology/access (013)	3 ISL levels $\times$ 4 GSL limits, 2 traffic regimes, 10 seeds, 3 phases; three non-probe policies, hash 0	720	781,230

fixed-topology states, not a continuous 5,838-second control-plane simulation. At 3,892 s one ground station is isolated and all four policies route 49 of the 50 flows in the orbit experiment. That phase is retained as a common access-geometry event.

B. Policy Admission and Inference

Util-LS parameters were selected using pre-deletion route and load diagnostics. The admission checks required at least 5% median path change from IGP-ECMP, bounded churn over the last three warm-up updates, and limited propagation and peak-stress inflation. Of three passing configurations, the recorded selection rule chose the one with the largest median path change, subject to fixed tie-breakers. No deletion outcomes were used for this selection. Across the subsequent 180 hash-scenario states, median path change was 0.14, median/p90 tail-three churn was 0.04/0.10, and the median propagation-RTT ratio to IGP-ECMP was 1.00022. Peak stress worsened in 14 of 180 states. Passing admission means the perturbation is usable for this comparison, not that the policy is an optimizer or has fully converged.

We aggregate scenario-level transfer statistics, not individual deletion rows. For the short-window and orbit comparisons, 10,000 bootstrap resamples use traffic-model/seed clusters, retaining all snapshots and relevant hashes within a draw. There are only 12 such traffic clusters, so these intervals describe the sampled workloads and should not be read as population-level guarantees. Hash-invariant policy pairs are counted only once. Undefined target-oracle cases are excluded from normalized regret and their counts are reported.

Analysis criteria were locked in repository experiment files before each corresponding formal analysis; this is internal registration, not an external preregistration claim. The short-window material-transfer criterion requires median $TR@20 \geq 0.20$, a 95% cluster-bootstrap lower bound above 0.10, recurrence in a strict majority of traffic seeds, and at least two snapshots. The orbit study additionally requires at least 12 of 24 phase medians to reach 0.20 and at least four of six confirmation phases to do so under at least two of three hashes. The directions are judged separately. These thresholds are practical analysis criteria, not a universal definition of non-transferability or multiple-testing-adjusted significance.

After the short-window result identified peak stress as the affected dimension, the orbit, traffic-factor, and topology studies locked that dimension as their primary outcome. Propagation, rerouting, and lost demand remain separate checks. We do not replace a failed primary criterion with a favorable secondary statistic.

C. Traffic and Topology Interventions

The traffic experiment crosses total offered load (25/100 units), concentration (equal versus 80% of demand on 20% of flows), flow count (25/100), and endpoint count (10/40). Its 16 regimes use ten traffic seeds at 0, 1,946, and 3,892 s. Endpoint pools and flow sequences are nested within each seed. Fixed ridge models predict Util-LS-to-shortest/IGP regret from policy labels, controlled factors, and pre-deletion features including path-incidence divergence, demand concentration, alternative-path diagnostics, and load-vector differences. Post-deletion outcomes and candidate identities are excluded from the inputs. Five-fold validation holds out two entire traffic seeds per fold. The locked criterion requires at least 10% relative MAE improvement over policy labels, a seed-cluster bootstrap interval above zero for mean error improvement, and improvement in at least three folds.

The topology experiment crosses three nested ISL layouts with GSL limits of 1, 2, 4, or 8 nearest-visible satellites. Sparse ISLs retain in-plane rings and every other adjacent-plane connection. The standard layout is the base plus-grid; the dense layout adds adjacent-plane diagonal alternatives. Their ISL counts are 1,734, 2,312, and 4,624, respectively. Every candidate link is screened by the geometric distance limit. The three layouts have satellite node connectivity 2, 4, and 8 at each tested phase.

This experiment uses two traffic regimes with the same 100-flow/40-endpoint OD structure: low/equal demand (25 units) and high/concentrated demand (100 units). They were chosen after the traffic-factor study to bracket load and concentration while holding flow and endpoint counts fixed, not as the observed minimum and maximum regret cells. Topology outcomes had not been inspected. The registered primary contrast is dense ISLs/GSL-8 minus sparse ISLs/GSL-1, paired by regime, seed, and phase. It tests *joint topology/access expansion*. Adding links also adds aggregate synthetic capacity; this is not a pure path-diversity intervention at fixed total capacity.

Transfer mitigation requires at least 75% defined pairs, median paired regret change at most -0.10 , a bootstrap upper bound below zero for the equally weighted mean of traffic-model/seed cluster means, and negative medians in at least seven seeds and two phases. Absolute missed positive damage, the oracle sum minus captured sum in Eq. 8, is also reported, assigning zero when the target oracle is zero. This keeps a change in the normalization denominator from being mistaken for a change in absolute damage.

V. RESULTS

A. Exposure Is Useful, but Is Not a Damage Oracle

The diagnostic experiment provides a useful correction to an exposure-only account. For positive propagation burden, same-policy exposure top-20 lists capture median fractions of 0.620, 0.617, and 0.500 of oracle damage under shortest-distance, hop-count, and the stress probe. Uniform selection from the same dependency universe gives 0.137, 0.128, and 0.157, respectively. These random values are medians across scenarios of each scenario’s median over 10,000 random sets. Exposure is informative, but it leaves appreciable damage uncaptured even before a policy change.

Deletion-defined rankings also transfer better than the exposure story suggests. Shortest-to-hop and hop-to-shortest median regret for positive propagation burden is only 0.030 and 0.037. The four directions involving the stress probe instead range from 0.415 to 0.458. This experiment alone would support a strong-perturbation result, not a general failure of ranking transfer between ordinary shortest-path-like policies.

B. Transfer Loss Is Direction- and Damage-Specific

Figure 1 shows the short-window peak-stress result for the three non-probe policies. Util-LS candidates evaluated under shortest-distance have median regret 0.398 (95% cluster-bootstrap interval $[0.251, 0.667]$); under IGP-ECMP the median is 0.412 ($[0.229, 0.667]$). Regret is defined in 162 and 165 of the 180 scenario-hash observations, respectively. Both directions meet the short-window criterion. Median regrets in the reverse directions are smaller, 0.173 and 0.155, and do not meet it. Shortest-distance and IGP-ECMP have median regret zero in both directions. A zero median does not imply every failure list transfers perfectly.

The result is not a general loss of failure predictability. Propagation burden, paired propagation stretch, rerouted demand, and lost demand do not meet the same non-probe transfer-loss criterion. Even the qualifying stress result varies with traffic: gravity traffic has median regret zero in both Util-LS source directions. Randomizing cutoff ties yields medians 0.392 and 0.414, close to the deterministic short-window values. The result therefore survives this tie check, but remains specific to the load dimension and workload scope.

C. Orbital Coverage Narrows the Claim Further

Figure 2 reports phase-level medians, and Table II separates the primary hash-0 summary from hash confirmation. Across the 24 phases, Util-LS-to-shortest regret has median 0.242

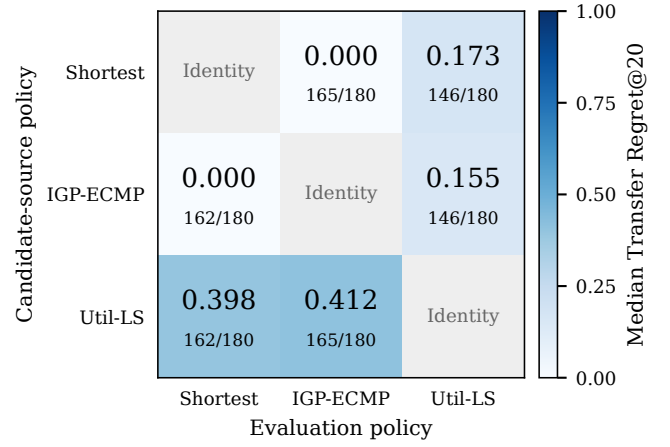


Fig. 1. Short-window median TR@20 for deletion-defined peak-link-stress rankings (EXP-RD-010). Rows select the list; columns evaluate it. Larger values mean more target-oracle damage missed. Counts give defined observations out of 180 for off-diagonal cells. Diagonals are normalization identities, not experimental evidence.

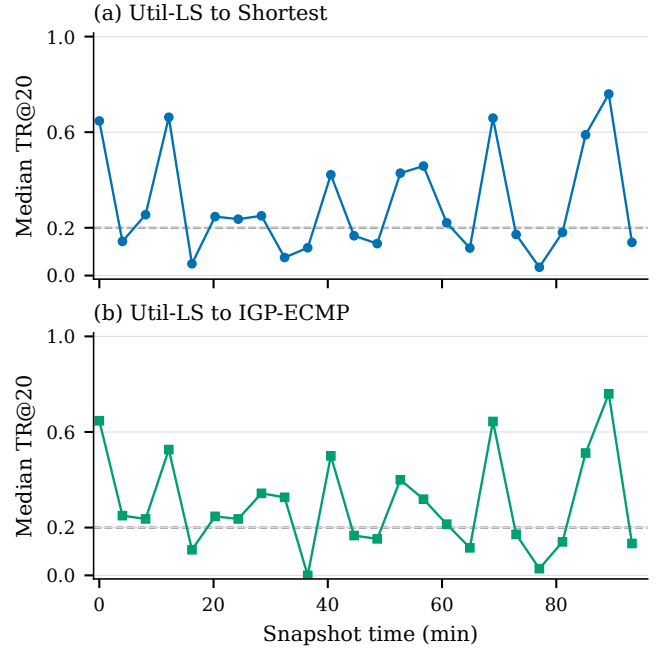


Fig. 2. Phase-wise median peak-stress TR@20 for Util-LS source lists, hash 0 (EXP-RD-012). Points represent independent warmed snapshots. The dashed line is the 0.20 material-regret threshold; crossing it at a phase is not sufficient to pass the separate three-hash confirmation in Table II.

and Util-LS-to-IGP regret has median 0.248. Both pass the primary full-orbit component. Only the IGP-ECMP direction passes the additional confirmation: four of six phases reach the threshold under a majority of hashes, versus two of six for shortest-distance. Thus we retain an orbit-sampled peak-stress result for IGP-ECMP and describe the shortest-distance result as phase/hash-sensitive. Reporting only the two pooled medians would obscure that difference.

TABLE II

ORBIT-SAMPLED PEAK-STRESS TRANSFER FROM UTIL-LS. INTERVALS ARE TRAFFIC-CLUSTER BOOTSTRAP INTERVALS FOR THE SCENARIO MEDIAN. CONFIRMATION COUNTS PHASES SUPPORTED BY AT LEAST TWO OF THREE HASHES; FOUR OF SIX ARE REQUIRED.

Target policy	Shortest	IGP-ECMP
Defined hash-0 observations	263/288	262/288
Median TR@20	0.242	0.248
95% interval	[0.148, 0.320]	[0.150, 0.333]
Phases with median ≥ 0.20	13/24	15/24
Hash-confirmed phases	2/6	4/6
Complete criterion	Not met	Met

Cross-time reuse can fail even without a policy change. At a one-bin phase separation (about 243 s), within-policy median peak-stress regret is 0.769 for shortest-distance, 0.800 for IGP-ECMP, and 0.781 for Util-LS. These diagnostics select candidates at the source phase without consulting the target phase; a candidate outside the target dependency universe has zero target damage. The large regrets show that a list tied to one geometry can lose relevance quickly in this sampled model. They do not establish a real-time refresh interval, because topology and routing state are initialized independently at each phase.

D. Traffic Features Do Not Yet Explain Transfer Reliably

The factorial experiment produces defined primary regret in 832 of 960 Util-LS-to-shortest/IGP comparisons. Under seed-held-out validation, policy labels alone give MAE 0.3112; adding the registered pre-deletion features reduces it to 0.2927. The 5.94% relative improvement falls short of the locked 10% criterion. Although all five folds improve in MAE, the seed-cluster bootstrap interval for mean absolute-error improvement is $[-0.0090, 0.0369]$. RMSE also worsens. We therefore do not present these features as a validated predictor or a causal explanation of transfer loss.

Among pairs with defined regret at both factor levels, higher offered load has an exploratory mean/median regret contrast of $+0.185/+0.083$. The concentration, flow-count, and endpoint-count contrasts have near-zero medians. This provides a reason to investigate offered load further, but does not rescue the failed predictive criterion. In particular, the 10-versus-40 endpoint result is not evidence that ranking transfer is insensitive to much denser real-world ground deployment.

E. More Links Do Not Restore Transfer in the Tested Comparison

The topology manipulation worked: edge sets were nested, the satellite graphs remained connected, and the dense/GSL-8 endpoint increased the median node-disjoint-alternative fraction from 0.00 to 0.24 relative to sparse/GSL-1. Nevertheless, the registered transfer-mitigation criterion failed for both evaluation policies.

Table III reports the paired comparison. Both directions have 51 of 60 pairs with positive target-oracle support at

TABLE III

DENSE ISLS/GSL-8 MINUS SPARSE ISLS/GSL-1 IN PEAK-STRESS TR@20. SOURCE POLICY IS UTIL-LS. ALL DIFFERENCES ARE DIMENSIONLESS. THE INTERVAL BELONGS TO THE EQUAL-CLUSTER MEAN, NOT THE PAIRED MEDIAN.

Target policy	Shortest	IGP-ECMP
Defined pairs	51/60	51/60
Median paired difference	+0.317	+0.132
Equal-cluster mean difference	+0.221	+0.150
95% interval for that mean	[0.071, 0.354]	[0.028, 0.269]
Negative seed medians	3/10	2/10
Negative phase medians	0/3	0/3
Mitigation criterion	Not met	Not met

both endpoints. Median regret *increases* by 0.317 for shortest-distance and 0.132 for IGP-ECMP. The bootstrap intervals for the equally weighted cluster means are wholly positive. No phase has a negative median contrast, and only three of ten seeds for shortest-distance and two for IGP-ECMP do. More alternatives therefore did not make these Util-LS critical lists more reusable under the registered joint expansion.

Normalized regret is not absolute damage. Across all 60 pairs, including zero-oracle cases, mean absolute missed positive stress damage changes by -0.00375 for shortest-distance and -0.03417 for IGP-ECMP. But paired medians are positive ($+0.0625$ and $+0.025$), and missed damage increases in 42/60 and 33/60 pairs. The mean decreases are therefore mixed distributional signals, not evidence of a consistent resilience improvement. The target oracle itself also changes as links are added. Neither a smaller oracle denominator nor a slightly lower mean absolute damage establishes better ranking transfer.

F. Integrity Checks

The formal outputs validate unique deletion keys, one-to-one flow pairing, finite values, demand-fraction bounds, and removal of the failed satellite from every post-failure route. The orbit bundle contains 822,996 deletion-policy rows; all 235,040 strict inactive controls have exactly zero impact. The topology bundle validates all 720 scenarios and 781,230 deletion-policy rows, along with graph and output hashes. These checks protect the comparison from several implementation errors, but do not validate the physical fidelity of the model. Ground-station access losses and unsupported delay tails remain explicit rather than being removed to improve the result.

VI. DISCUSSION AND LIMITATIONS

A. What Should Be Reused?

The main lesson is not that every routing change invalidates a criticality report. Some rankings transfer well, and the observed failure is narrower than our initial exposure comparison suggested. A useful report should name the damage dimension, routing and failure-response state, traffic instance, and time at which its ranking was evaluated. Damage Capture then asks a practical question: how much would an analyst miss by reusing

that list? This also makes stable transfer a meaningful result instead of treating every ranking difference as a vulnerability.

Our strongest confirmed exception concerns peak-link-stress damage from Util-LS lists evaluated under IGP-ECMP. It does not establish that Util-LS improves congestion, that operational routing is unsafe, or that delay and reachability rankings fail in the same way. The failed traffic predictor and topology-mitigation criterion leave the mechanism unresolved. A follow-up must distinguish changes in bottleneck placement, target-oracle magnitude, access alternatives, and total capacity, rather than label their combination a single path-diversity effect.

B. Model and External Validity

The evaluation is graph-level simulation with synthetic demand and uniform capacity. It omits packet queues, transport feedback, heterogeneous link budgets, directional capacity, pointing constraints, weather, gateway backhaul, and operator link scheduling. Peak stress is therefore a controlled load-concentration measure, not measured utilization or packet-level congestion. The propagation RTT estimate and modeled congestion-adjusted RTT remain distinct from operational RTT. No ns-3 packet-level validation is reported in this revision.

IGP-ECMP and Util-LS add mechanisms beyond distance minimization, but neither is a deployed LEO routing system. Util-LS retains residual warm-up churn, freezes advertisements during failure evaluation, and is reinitialized at each phase. Continuous topology/control-plane interaction, detection time, and convergence transients remain untested. Hash-0 traffic and topology studies also have a narrower scope than the three-hash policy and orbital checks. A finite set of internally locked thresholds does not remove this model dependence.

The 100 ground locations are a controlled geographic input. Varying active endpoints or GSL multiplicity does not establish realism for thousands of terminals and gateways, nor for an operator's traffic matrix. The topology intervention changes aggregate capacity as well as routing choices. Its negative mitigation result applies to the tested two regimes and three phases, not every dense constellation.

An earlier exploratory experiment projected a deterministic top-400 subset of public Starlink TLEs onto a synthetic grid. That construction can mix orbital groups and bias geometry, so it is not used here as external validation of the revised deletion-defined claim. A corrected, shell-aware, physically screened fixed-TLE study with multiple subset seeds is still pending. Real orbital elements alone would not reveal actual ISLs, traffic, or routing. Measurements of deployed Starlink performance illustrate why geometric models should not be equated with service behavior [9].

Finally, the current random baseline does not replace service-aware centrality and structure-matched selectors. Those comparisons must obey the same no-ground-relay rule and candidate universe. The legacy full-graph centrality overlap does not satisfy that comparison and is not retained as evidence of superiority. Likewise, our single-failure rankings

do not characterize simultaneous or cascading failures. These are substantive remaining evaluations, not results implied by the number of completed runs.

VII. RELATED WORK

Topology and time. Hypatia supports reproducible satellite-network studies using orbital geometry and time-varying graphs [5]. Topology design work examines the latency/capacity tradeoffs of different ISL structures [1], while route-variability studies characterize path churn and RTT variation [2]. Handley's low-latency routing study likewise demonstrates the importance of geometry and path choice [10]. We use this modeling foundation to examine reuse of deletion rankings rather than optimize the constellation or its paths. The broader temporal-network literature establishes that node importance and dynamics can depend on time [11]; our cross-time test asks how much failure damage a list from one phase captures at another.

Critical nodes and load dependence. The closest prior satellite-resilience study evaluates the effect of removing highly used satellites under shortest-path routing [3]. We retain deletion-and-rerouting as the intervention, but evaluate all registered policy-dependent satellites and separate source ranking from target evaluation. Classical betweenness methods provide another family of predictors [12]. HYDRA's Hyper-Bridge Centrality uses load and redundancy within dependency structures to identify vulnerabilities [4]. Thus, load-dependent risk is not our novelty claim. Our focus is the transfer of a dimension-specific deletion ranking, normalized by the target's own oracle. We do not claim empirical superiority over centrality or HYDRA without a matched comparison.

Routing and emulation. ECMP hashing distributes flows across equal-cost alternatives [6], and more general link-state forwarding schemes can support traffic engineering [13]. Our Util-LS model is simpler and carries no optimality claim. StarryNet provides protocol-oriented space/terrestrial experimentation [14]; such an environment could test consequences omitted by the present graph model. We choose the lighter abstraction to enumerate failure effects across policies and scenarios, not because it reproduces packet-level behavior.

VIII. CONCLUSION

Different routes expose different satellites, but that observation does not establish that a failure ranking has become unusable. We test reuse against deletion-defined target-policy damage, with paired flows, a common dependency universe, and explicit zero-support handling. Shortest-path-like rankings often transfer well. The confirmed exception in our controlled shell is a peak-stress ranking selected under Util-LS and evaluated under IGP-ECMP across sampled orbital phases. The shortest-distance direction is less stable across phases and hashes, the registered traffic predictor does not generalize sufficiently, and adding ISL/GSL alternatives does not restore transfer in the tested contrast. The evidence favors conditional, damage-specific validation of ranking reuse over either a

universal critical-satellite list or a universal claim of routing-induced instability.

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REFERENCES

- [1] D. Bhattacharjee and A. Singla, “Network topology design at 27,000 km/hour,” in *Proc. ACM CoNEXT*, 2019, pp. 341–354, doi: 10.1145/3359989.3365407.
- [2] V. Bhosale, A. Saeed, K. Bhardwaj, and A. Gavrilovska, “A characterization of route variability in LEO satellite networks,” in *Passive and Active Measurement*, 2023, pp. 313–342.
- [3] Z. Li, W. M. Zhang, W. Chen, Y. Hu, and W. Lu, “LEO satellite network resilience analysis: A focus on critical satellites,” in *Proc. 2nd International Workshop on LEO Networking and Communication*, 2024, pp. 13–18, doi: 10.1145/3697253.3697267.
- [4] B. Yuan, M. Tang, B. Ge, H. Luo, and Z. Yan, “HYDRA: Unearthing ‘black swan’ vulnerabilities in LEO satellite networks,” arXiv:2602.06612, 2026.
- [5] S. Kassing, D. Bhattacharjee, A. B. Águas, J. E. Saethre, and A. Singla, “Exploring the ‘Internet from space’ with Hypatia,” in *Proc. ACM Internet Measurement Conference*, 2020, pp. 214–229, doi: 10.1145/3419394.3423635.
- [6] C. Hopps, “Analysis of an equal-cost multi-path algorithm,” RFC 2992, Nov. 2000, doi: 10.17487/RFC2992.
- [7] Kuiper Systems LLC, “Technical Appendix: Kuiper System,” FCC application technical appendix, 2019. [Online]. Available: <https://cdn.geekwire.com/wp-content/uploads/2019/07/Kuiper-Technical.pdf>.
- [8] Natural Earth, “Populated places,” [Online]. Available: <https://www.naturalearthdata.com/downloads/10m-cultural-vectors/10m-populated-places/>.
- [9] F. Michel, M. Trevisan, D. Giordano, and O. Bonaventure, “A first look at Starlink performance,” in *Proc. ACM Internet Measurement Conference*, 2022, pp. 130–136, doi: 10.1145/3517745.3561416.
- [10] M. Handley, “Delay is not an option: Low latency routing in space,” in *Proc. ACM HotNets*, 2018, pp. 85–91.
- [11] P. Holme and J. Saramäki, “Temporal networks,” *Physics Reports*, vol. 519, no. 3, pp. 97–125, 2012, doi: 10.1016/j.physrep.2012.03.001.
- [12] U. Brandes, “A faster algorithm for betweenness centrality,” *Journal of Mathematical Sociology*, vol. 25, no. 2, pp. 163–177, 2001, doi: 10.1080/0022250X.2001.9990249.
- [13] D. Xu, M. Chiang, and J. Rexford, “Link-state routing with hop-by-hop forwarding can achieve optimal traffic engineering,” *IEEE/ACM Transactions on Networking*, vol. 19, no. 6, pp. 1717–1730, 2011.
- [14] Z. Lai, H. Li, Y. Deng, Q. Wu, J. Liu, Y. Li, J. Li, L. Liu, W. Liu, and J. Wu, “StarryNet: Empowering researchers to evaluate futuristic integrated space and terrestrial networks,” in *Proc. USENIX NSDI*, 2023, pp. 1309–1324.